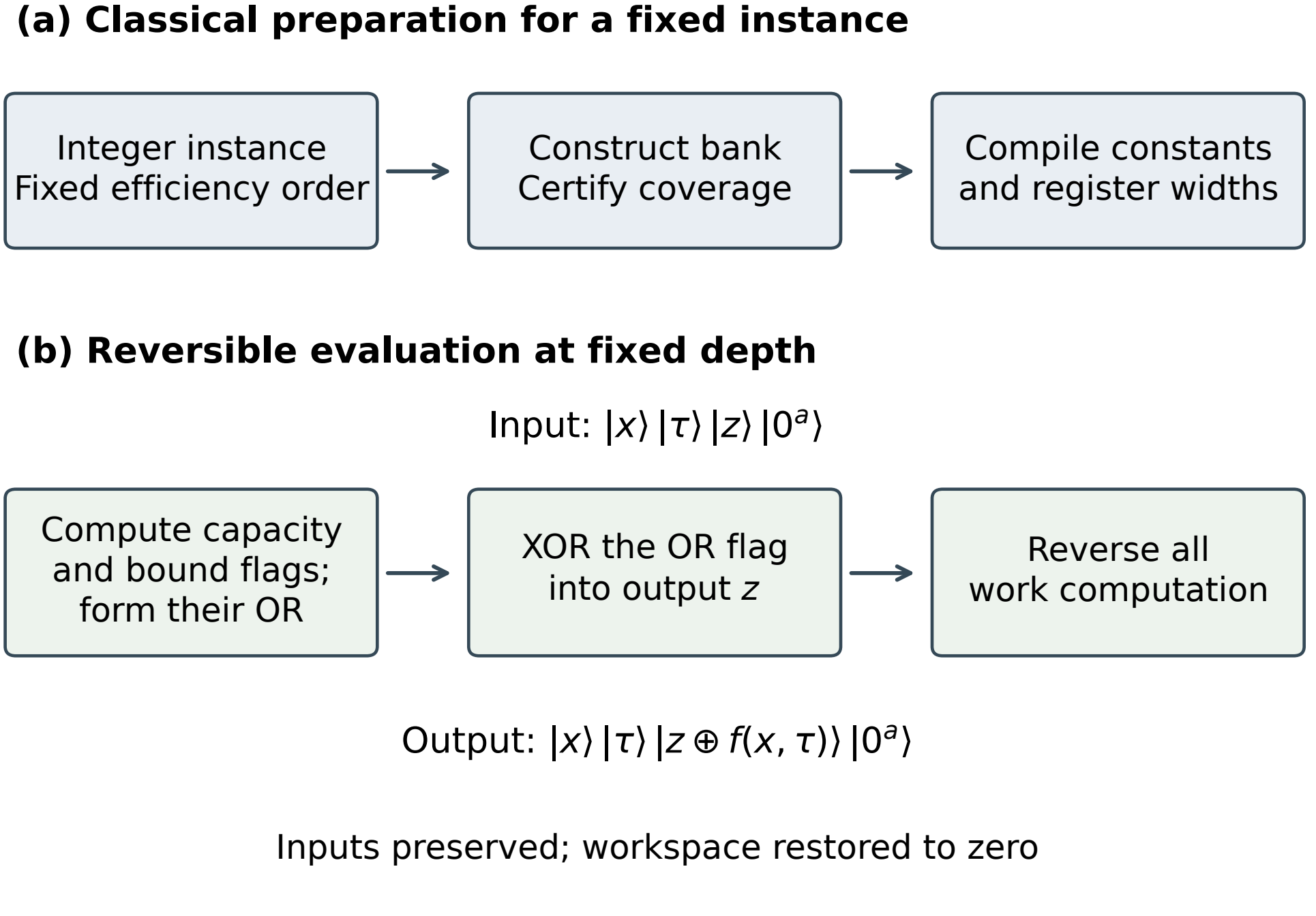
(a) Classical preparation for a fixed instance
Integer instance
Fixed efficiency order
Construct bank
Certify coverage
Compile constants
and register widths
(b) Reversible evaluation at fixed depth
Input: $|x\rangle\,|\tau\rangle\,|z\rangle\,|0^a\rangle$
Compute capacity
and bound flags;
form their OR
XOR the OR flag
into output z
Reverse all
work computation
Output: $|x\rangle\,|\tau\rangle\,|z \oplus f(x,\tau)\rangle\,|0^a\rangle$
Inputs preserved; workspace restored to zero